Wednesday Warm Up: Unit 4, Inductors in Series and Parallel, and RL circuits

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April 1, 2026

1 Memory Bank

- $\epsilon = -N\Delta\phi_m/\Delta t$ ... Faraday's Law
- $\epsilon = -L\Delta I/\Delta t$ ... Faraday's Law
- 1 H $= 1$ Ω s ... The *Henry* is a unit of inductance, often represented by L.
- $\mu_0 = 4\pi \times 10^{-7}$ ... T m^{-1} A^{-1}.
- $L = \mu_0 A N^2/l$ ... Inductance of a solenoid.
- $X_{\rm C} = 1/(2\pi f C)$ ... Reactance of a capacitor.
- $X_{\rm L} = 2\pi f L$ ... Reactance of an inductor.

2 Inductors in Series and Parallel

1. **Inductors in Series.** Show that the total inductance $L_{\rm tot}$ of two inductors with inductances L_1 and L_2 in *series* is
$$L_{\rm tot} = L_1 + L_2 \tag{1}$$
(a) Draw a diagram below. (b) Is the current the same through both L_1 and L_2? (c) Write a formula relating the voltage across each inductor and the voltage across both inductors. (d) Substitute Faraday's law to prove Eq. 1.

2. **Inductors in Parallel.** Show that the total inductance $L_{\rm tot}$ of two inductors with inductances L_1 and L_2 in *parallel* is
$$L_{\rm tot}^{-1} = L_1^{-1} + L_2^{-1} \tag{2}$$
(a) Draw a diagram below. (b) Write a formula relating the current through each inductor and the total current, and then take the derivative of both sides ($\Delta I/\Delta t$). (c) Is the voltage the same across both L_1 and L_2? (d) Substitute Faraday's law to prove Eq. 2.

3. What rate of current increase will give a back emf of 20 mV for two 100 μH inductors in series?

4. Camera flashes charge a capacitor to high voltage by switching the current through an inductor on and off rapidly. In what time must the 0.100 A current through a 2.00 mH inductor be switched on or off to induce a 500 V emf?

3 LC Circuits

1. Recall that a charged LC circuit oscillates with frequency $f = 1/(2\pi\sqrt{LC})$. (a) If $L = 100$ μH, and $C = 100$ μF, what is the angular resonance frequency? (b) What is the regular frequency, in Hz? (c) Sketch the AC signal below, if it has an amplitude of 5 V.

2. The reactance of a capacitor is $X_{\rm C} = 1/(2\pi f C)$, and the reactance of an inductor is $X_{\rm L} = 2\pi f L$. Set these equal, and solve for the frequency. What do you find?